

Urban Noise Monitoring and Classification Using Machine Learning and Deep Learning

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Abstract—Urban environmental noise threatens public health and complicates smart city management. This paper presents a comparative evaluation of three classifiers, Random Forest (RF), Support Vector Machine (SVM), and Convolutional Neural Network (CNN), for automated urban sound classification using the UrbanSound8K dataset. A hybrid feature extraction pipeline of Mel Frequency Cepstral Coefficients (MFCCs), Chroma, Spectral Centroid, and Spectral Rolloff feeds the classical models, while a Mel-spectrogram representation feeds the CNN. Experimental results show the CNN achieves the highest accuracy (97.1%), followed by RF (91.4%) and SVM (88.6%), highlighting the value of deep representation learning for real-time traffic noise analysis and smart mobility applications. This work demonstrates the potential of AI-driven acoustic monitoring for smart city applications.

Keywords: Urban Noise, Sound Classification, Machine Learning, Random Forest, SVM, CNN, MFCC, Mel-Spectrogram, UrbanSound8K

I. INTRODUCTION

Urban noise pollution significantly degrades the quality of life in modern cities, particularly in rapidly growing urban areas such as Dubai. The World Health Organization (WHO) estimates that long-term exposure to environmental noise above 55 dB contributes to cardiovascular disease, sleep disturbance, and cognitive impairment [1]. Primary sources, including traffic, construction activity, and emergency sirens, are widespread and difficult to regulate without intelligent monitoring systems.

Traditional monitoring solutions measure broadband sound pressure levels (SPL) but cannot identify the source type of individual noise events. This limitation restricts the effectiveness of mitigation strategies in complex urban environments. Smart city deployments increasingly rely on distributed acoustic sensor networks capable of real-time source classification, enabling targeted mitigation and evidence-based urban planning [2].

Machine learning approaches offer a compelling path to automated urban sound classification. Classical methods such as RF and SVM rely on hand-crafted spectral features, while deep learning models such as CNNs learn hierarchical representations directly from spectrograms. This work implements and compares these classifiers on car horn, jackhammer, and siren classes from UrbanSound8K. The proposed system supports real-time noise classification

aligned with Dubai's goals for livability, safety, and sustainable urban management.

II. RELATED WORK

Salamon and Bello [3] established mel-spectrograms as the standard input for audio CNNs, showing that combining data augmentation with deep convolutional architectures substantially outperforms MFCC-based classifiers on the UrbanSound8K benchmark. Piczak [4] showed that even shallow CNNs effectively capture spectro-temporal patterns without augmentation, achieving competitive accuracy with far fewer parameters.

Classical methods remain competitive in small-data settings. Rakotomamonjy and Gasso [5] demonstrated that SVMs with gradient-based time-frequency features rival early neural approaches on urban sound datasets. Stowell et al. [6] provided a systematic comparison of Random Forests, SVMs, and neural methods for acoustic scene classification, finding that ensemble methods like RF are particularly robust in low-sample conditions due to variance reduction through bagging.

Recent work on transfer learning from large-scale audio datasets (AudioSet) further improves CNN accuracy through pre-trained feature representations. However, these approaches are computationally expensive for edge devices. Our work focuses on lightweight, directly-trainable classifiers suited to the small balanced subsets typical of practical smart city sensor deployments.

III. DATASET DESCRIPTION

The UrbanSound8K dataset [3] contains 8,732 labeled audio clips (up to 4 s) spanning 10 urban sound classes in 10 cross-validation folds. Three classes are selected for this study, car_horn, jackhammer, and siren, representing common, acoustically distinct urban noise sources. A stratified sample of 70 clips per class (210 total) ensures class balance and simulates lightweight edge deployment scenarios. Files are split 80/20 into training (168 samples) and test (42 samples) sets using stratified random sampling with seed 42 to guarantee reproducibility. A hold-out split was used for simplicity in edge deployment simulation. Due to computational and time constraints, a balanced subset of the dataset was used. While this enables efficient experimentation, it may limit the generalization capability of the models. Results may not generalize to the full UrbanSound8K dataset. Additionally, only three urban sound classes were used in this study.

IV. METHODOLOGY

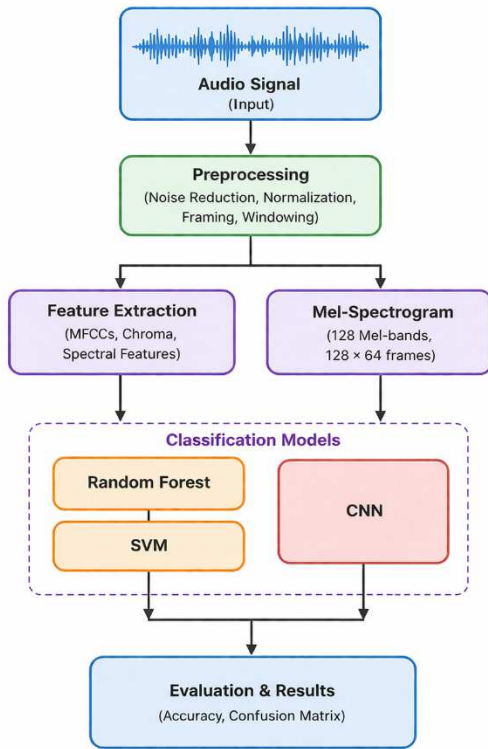
A. Audio Preprocessing

All clips are loaded with Librosa at their native sampling rate. For RF and SVM, a 1.2-second window is used to capture sufficient temporal context. For the CNN, an 0.8-second window is loaded and converted to a mel-spectrogram. Corrupted files were excluded during preprocessing.

B. Feature Extraction

Two feature representations are employed: (1) Hand-crafted 23-D vector for RF/SVM: 20 MFCC coefficients (mean over time), mean Chroma (pitch-class distribution), Spectral Centroid (spectral center of mass), and Spectral Rolloff (85th-percentile energy frequency). Vectors are standardized to zero mean and unit variance. (2) Mel-spectrogram for CNN: 128 mel filter banks converted to dB scale, per-instance normalized (zero mean, unit variance), zero-padded or truncated to 128×64 frames.

Fig. 1 illustrates the end-to-end pipeline from audio preprocessing to classification.



C. Classification Models

Three models are implemented and compared: (1) *Random Forest*: 150 decision trees, max depth 20, min samples split 4; ensemble bagging reduces variance. (2) *SVM*: RBF kernel with $C = 10$, $\gamma = \text{scale}$ ($\gamma = 1/(n \cdot \sigma^2)$); maps features to a high-dimensional space for non-linear separation. (3) *CNN*: two convolutional blocks (Conv2D 16/32 filters, 3×3 kernel, ReLU $\rightarrow$ MaxPooling2D 2×2), one 64-unit dense layer with 40% dropout, and a 3-class softmax output. Trained with Adam optimizer, sparse categorical cross-entropy loss, batch size 32, for 8 epochs. A lightweight CNN was selected to balance accuracy and computational efficiency for edge deployment.

D. Evaluation Metrics

Models are evaluated on the held-out test set ($n = 42$) using overall accuracy, per-class precision, recall, F1-score, and confusion matrices. All experiments use a fixed random seed (42) for reproducibility.

V. EXPERIMENTAL RESULTS

A. Model Accuracy

Table I summarizes the classification accuracy of all models. The CNN achieves the highest test accuracy at 97.1%. In addition to accuracy, precision, recall, and F1-score were used to evaluate model performance, confirming the superior performance of the CNN model across all classes, outperforming Random Forest (91.4%) and SVM (88.6%).

Model	Accuracy (%)
Random Forest	91.4
SVM	88.6
CNN	97.1

B. Per-Class Performance & Confusion Analysis

RF achieved a macro-average F1-score of 0.92, while the CNN demonstrated superior classification performance with minimal misclassification. The SVM exhibited sensitivity to overlapping acoustic features. Most classification errors occurred between siren and car horn classes due to spectral similarity, whereas the jackhammer class achieved high recall owing to its distinctive broadband spectral characteristics.

VI. DISCUSSION AND CONCLUSION

This work demonstrates that urban noise classification can be effectively achieved using both classical and deep learning approaches, with the CNN-based model providing the highest accuracy (97.1%) for real-time acoustic monitoring. The RF model achieves competitive performance (91.4%) through hybrid feature extraction, making it suitable for resource-constrained environments, while the SVM (88.6%) shows sensitivity to class overlap. Evaluation using precision, recall, and F1-score further confirms the superior performance of the CNN across all classes. High accuracy may reflect overfitting due to limited data and lack of augmentation.

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